

BDS BOOTCAMP WEEK

Session 3: AI & Research Tools

AI Assistants · Hugging Face · Obsidian · LaTeX

WELCOME BACK

How today works

- Last session of bootcamp week — a bit more talk, a bit less install-and-type
- Today: the tools that will help you work faster and think more clearly for the next two years
- We'll close with an open Q&A before your first real BDS lecture on Wednesday

TODAY'S AGENDA

- AI coding assistants
- Hugging Face
- Obsidian for research
- LaTeX for writing
- Recap of the whole week
- Open Q&A

PART 1

AI Coding Assistants

What's out there

- Two broad categories, that's really all you need to remember today
- General assistants (Claude, ChatGPT) — a chat window, good for writing and explaining code
- Coding-specific tools (GitHub Copilot, and more advanced ones like Codex or Claude Code) — live inside your editor or terminal
- You'll use these constantly in this program — the goal today is just knowing they exist and roughly what each kind is for

Editor with AI added: VS Code vs. Cursor

Cursor is a fork of VS Code — same look and keybindings, AI is just baked in natively.

VS Code (what we use)

- No AI out of the box — you add it via an extension (Copilot, Claude, etc.)
- Free, extremely widely used, huge extension ecosystem
- What we install and standardize on in Session 1

Cursor

- Chat, multi-file edits and autocomplete built natively into the editor
- Free tier, paid for heavier AI usage
- Worth knowing it exists — not required, and we won't teach it directly so everyone's setup matches

A more advanced kind: coding agents

One level beyond the assistants above: you hand off a task and it works through it for you. Not something you need to try this week — just good to know it exists.

Codex

- OpenAI's version
- Lives in a terminal or the cloud, edits files and runs commands for you

Claude Code

- Anthropic's version
- Same idea, same core ability — just a different company behind it

Don't stress over which one — they're close enough that it mostly comes down to what's provided or allowed.

Great for vs. be careful with

Great for

- Explaining an error message you don't understand
- Brainstorming different approaches to a problem
- Writing repetitive boilerplate code
- Learning the syntax of a new library quickly

Be careful with

- Trusting generated code without reading and testing it
- Using it to skip understanding the fundamentals
- Submitting AI-written work as your own where that's against course policy
- Feeding it sensitive/private data

Using them well

- Treat it like a very fast, occasionally wrong assistant — always verify what it gives you
- Ask it to explain code, not just generate it, if the goal is to actually learn
- Good prompts include context: what you're trying to do, what you already tried, what went wrong
- Check the specific policy for each course on AI use in graded work — it can vary by course

PART 2

Hugging Face

The "GitHub" of machine learning

- huggingface.co — a hub for pretrained models, datasets, and live demos
- Models: use a pretrained model instead of training one from scratch
- Datasets: thousands of ready-to-use datasets for practice and projects
- Spaces: live demo apps other people have built and shared

FOR NOW

Just get familiar
with the site.

You'll use it for
real later in the

Quick look

- Go to huggingface.co/datasets
- Search for a topic you're interested in (sports, finance, social media, ...)
- Open one dataset card and skim its description — that's the whole workflow

PART 3

Research & Writing Tools

Note-taking that mirrors how ideas connect

- A note-taking app built around linking ideas together, not folders and files
- Great for digesting papers: one note per paper or concept, linked to related ones
- Notes are plain local Markdown files — you own your data, works fully offline
- Not required for this course, but many students find it invaluable during thesis literature review

Getting started

- A "vault" is just a folder of notes
- Link one note to another with `[[double brackets]]` — over time this builds a web of ideas
- Suggested workflow for a paper: one note with your summary, linked to a broader "themes" note
- Graph view shows you visually how your notes connect — useful once you have a few dozen

Why LaTeX over Word

- Word fights you as documents get longer — formatting drifts, citations get messy
- LaTeX: you describe the structure in text, and formatting is generated consistently
- Handles references/citations, equations, and figures automatically and reliably
- Steeper learning curve up front — pays off hugely for your thesis and long reports

Getting started with Overleaf

- overleaf.com — LaTeX in the browser, nothing to install
- Real-time collaboration, like Google Docs but for LaTeX
- Start from a template — search "report" or "thesis" in Overleaf's template gallery
- Check whether Aalborg University has an official thesis LaTeX template — worth using if so

PART 4

Wrapping Up the Week

What I wish I knew when I started

- Ask questions early — confusion compounds if you sit on it
- Start using version control from day one, even for solo assignments
- Build your note-taking habit before your thesis forces you to, not during it
- The specific tool matters less than using some tool consistently

RECAP

Your toolkit after this week

- Terminal & file navigation
- Python via Anaconda
- VS Code + extensions
- Virtual environments (venv)
- Python basics: a first taste (more very soon)
- Jupyter notebooks & Google Colab (quick look)
- Git & GitHub, branches, pull requests
- AI coding assistants, used responsibly
- Hugging Face ecosystem
- Obsidian & LaTeX for research and writing

Looking ahead

- Wed 9 Sept — your first official BDS lecture
- Christian, Andreas & Roger stay on as your TAs throughout the semester
- A few short follow-up sessions later in September, once real coursework surfaces real questions
- Everything from this week stays available as reference material

NOW

Open Q&A

Ask anything about the program, the tools, or what's coming next.

That's bootcamp week — well done!

See you Wednesday for the first BDS lecture.